

Innovation Insight: AI Observability

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Initiatives: Delivery of Functional Responsibilities

AI systems introduce risks that traditional observability cannot detect, from hallucinations and bias to compliance violations. This research defines AI observability and gives heads of I&O the insights needed to evaluate, instrument, and govern AI systems in production.

Overview

Key Findings

- AI systems, which are inherently probabilistic, fail in ways that traditional, deterministic IT observability cannot detect. Hallucinations, model drift, bias, and output quality degradation produce no signal in infrastructure dashboards or application logs, leaving organizations exposed to risks that only purpose-built AI observability tooling can surface.
- AI observability benefits multiple business domains, including legal, compliance, privacy, audit, and IT security, by providing the visibility and evidence each requires to manage AI risk effectively. Clear governance and ownership structures are a prerequisite for realizing this value.
- Agentic AI introduces a qualitatively different observability challenge. Governing autonomous multistep AI workflows requires trace-level visibility across entire reasoning chains.
- The business case for AI observability extends well beyond risk mitigation. Outcome-based metrics demonstrate ROI, token and inference visibility make AI costs manageable, and comprehensive audit trails satisfy the governance requirements of emerging regulatory frameworks.

Recommendations

- Start AI observability at the application and inference layers, where risk signals are most visible, then expand coverage systematically across the full AI technology stack.
- Establish clear ownership and governance for AI observability before instrumentation begins, ensuring legal, compliance, privacy, and security teams are active participants from the outset.
- Evaluate AI observability tooling for both runtime monitoring and enforcing systematic evaluation through runtime guardrails, prioritizing open standards such as OpenTelemetry and modular architectures that can accommodate a rapidly evolving vendor landscape.
- Connect AI observability to business outcomes by defining outcome-based KPIs, establishing visibility into token consumption and inference costs, and ensuring observability produces the audit trails and governance evidence regulators and stakeholders require.

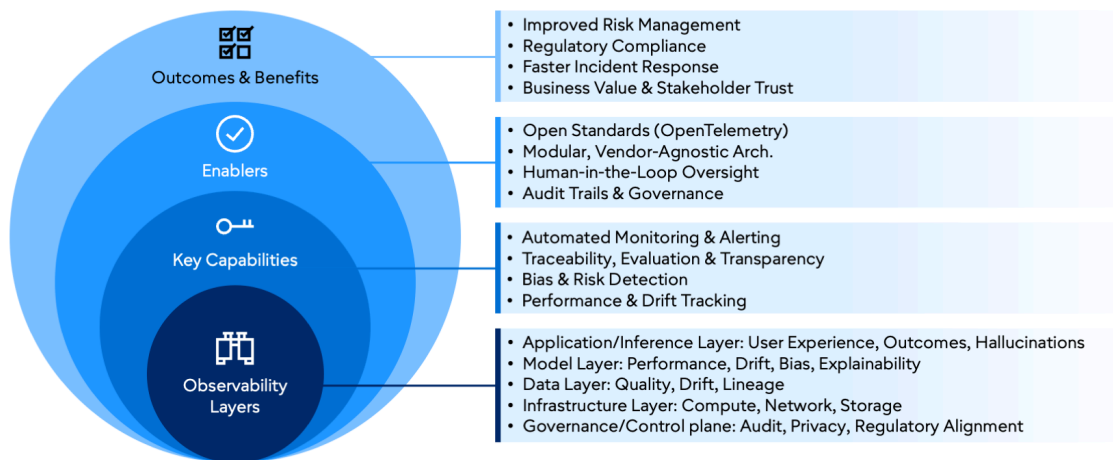
Introduction

Organizations are deploying AI faster than they can govern it. Traditional monitoring and observability platforms were not designed for nondeterministic AI systems, and the failure modes they miss — like hallucinations, bias, drift, and compliance violations — can cause financial and reputational damage before anyone knows something has gone wrong.

AI observability closes this gap. It gives heads of I&O the instrumentation, evaluation, and governance capabilities needed to operate AI systems with confidence. Figure 1 shows how the key layers and capabilities of AI observability combine to deliver risk management, regulatory compliance, and measurable business value.

Figure 1: AI Observability Blueprint: Key Layers, Capabilities, and Trustworthy Outcomes

AI Observability Blueprint: Key Layers, Capabilities, and Trustworthy Outcomes



Source: Gartner
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Description

Definition

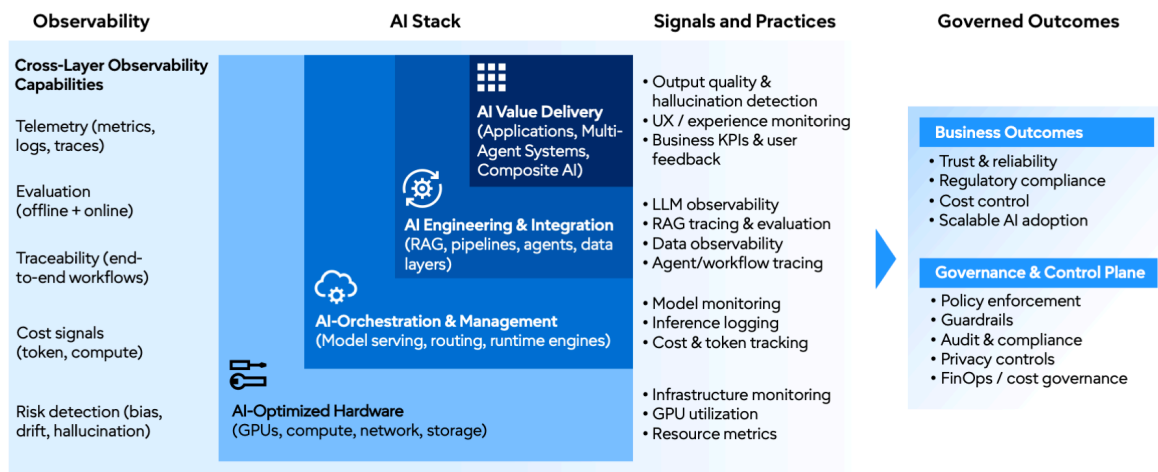
AI observability is the practice of collecting, analyzing, and acting on telemetry signals across all layers of an AI system to monitor behavior, diagnose failures, and drive continuous improvement. It extends traditional IT observability to address the nondeterministic nature of AI systems, where behavior emerges from the interaction of models, data pipelines, retrieval systems, and agentic workflows. AI observability combines automated instrumentation with human oversight. Together, these enable organizations to detect bias, drift, and compliance violations, and manage cost and accountability over AI systems in production.

Monitoring the AI Technology Stack With Layered Observability

AI systems are composed of multiple interconnected layers, each producing distinct signals and introducing its own failure modes. Because a problem at one layer can propagate silently to produce a poor outcome at another layer, no single observability product or vantage point is sufficient. Effective AI observability requires coverage across the full stack, with instrumentation appropriate to what each layer can and cannot reveal (see Figure 2).

Figure 2: AI Observability Across the Full Stack: From Infrastructure Signals to Business Outcomes

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- **Infrastructure Layer:** Tracks standard metrics like compute utilization, GPU memory, etc. AI workloads require dynamic thresholds to handle bursts in inference demand that traditional static monitoring might miss.
- **Model Layer:** Focuses on behavioral signals, including output distribution drift, accuracy degradation, and hallucinations. These require specialized tooling, often using OpenTelemetry, since traditional application performance monitoring (APM) is insufficient here.
- **Engineering Tools Layer:** Monitors the health of data pipelines, RAG retrieval components, and orchestration frameworks like LangChain. Traceability here is essential for root cause analysis of quality issues.
- **Application and Interface Layer:** Converges with digital experience monitoring (DEM) to track user-centric signals alongside AI-specific metrics like guardrail trigger frequency and token consumption.

Runtime telemetry (traditional observability signals) cannot confirm if an AI system is performing well. Systematic evaluation using human feedback and adversarial probing is required to surface risks and issues that live monitoring misses, ensuring systems are not just fast but also consistently right.

Benefits and Uses

Why Organizations Must Prioritize AI Observability

AI systems fail differently from traditional software. A misconfigured application typically produces an error that is visible, traceable, and reproducible. An AI system can fail silently. It can generate confident, plausible-sounding outputs that are biased or factually wrong, but with no corresponding alert in an infrastructure dashboard or application log. By the time the failure surfaces, the damage (to customers, reputation or regulatory standing) may already be done.

The case for AI observability is not purely defensive. Organizations that instrument their AI systems effectively gain capabilities that directly accelerate AI adoption and business value.

Visibility into model behavior builds the internal confidence needed to move AI initiatives from pilot to production at scale. Outcome-based metrics and KPIs give leaders the evidence needed to demonstrate ROI to boards and business stakeholders. Granular visibility into token consumption, compute usage, and inference costs transforms AI from an unpredictable cost center into a manageable, optimizable resource. The governance evidence that AI observability produces increasingly determines whether organizations can demonstrate that their AI systems are trustworthy.

Infrastructure monitoring and observability platforms cannot provide any of this. They were designed for deterministic systems where anomalies show up as measurable deviations in latency, error rates, or resource utilization. AI system degradation (drift, bias, hallucination, etc.) leaves no trace in infrastructure metrics. AI system qualities exist on largely independent axes: optimizing for speed, cost, and reliability does not ensure correctness, fairness, or legal compliance. Conversely, even accurate and well-governed systems may be slow or resource-intensive.

For heads of I&O, AI observability is the foundational capability that makes AI governable, scalable, and demonstrably valuable. Without it, managing AI risk is reactive, proving AI value is difficult, and scaling AI with confidence is not possible.

Governing Agentic and Autonomous Systems

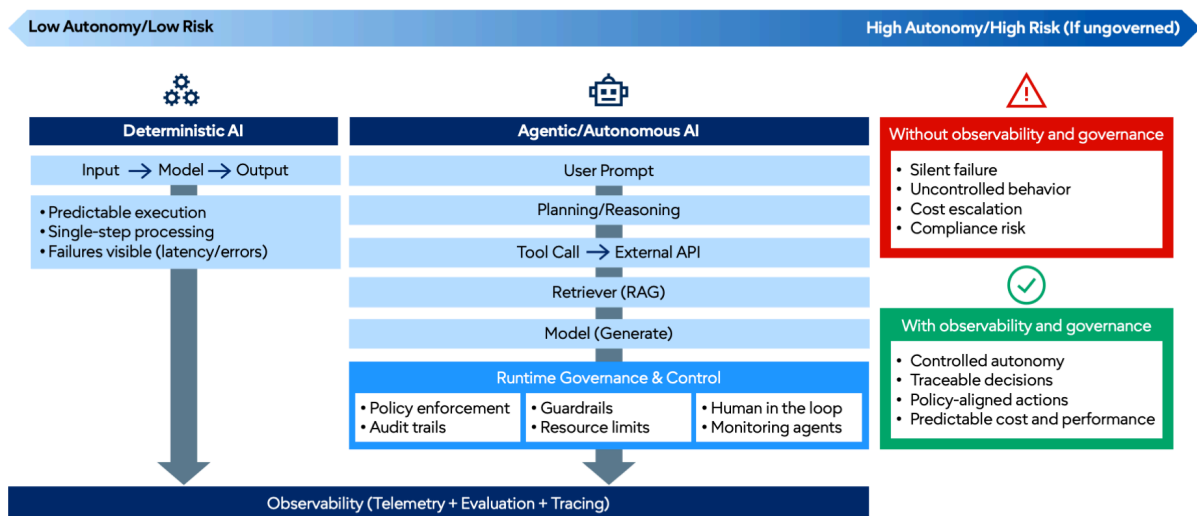
Agentic AI systems, which chain together models, tools, and decision steps to complete complex tasks with minimal human intervention, represent the fastest-growing and least-governed frontier of enterprise AI deployment. Unlike single-inference model calls, agentic workflows execute multistep reasoning chains, invoke external tools and APIs, retrieve and synthesize information from multiple sources, and in most configurations spawn additional agents to complete subtasks. Each step introduces its own failure mode, and errors compound across the chain in ways that are difficult to detect and harder to attribute.

Observing agentic systems requires a fundamentally different approach to instrumentation. Trace-level visibility across entire reasoning chains is necessary for understanding what an agentic system actually did and why. Without it, a flawed decision or a harmful output cannot be reliably traced back to its origin, making both remediation and accountability impossible. Capturing inputs, intermediate steps, tool calls, retrieved context, and final outputs is essential for providing this level of visibility and insight.

Governance adds a further dimension (see Figure 3). As agentic systems become more autonomous, the risk of ungoverned behavior increases. This includes systems that exceed their intended scope, consume disproportionate resources, or interact with external services in ways that were not anticipated at deployment. Policy enforcement at runtime, automated guardrails, and the use of dedicated monitoring agents to validate agentic behavior are emerging responses to this challenge, but comprehensive strategies for governing agentic AI at scale remain nascent across the industry.

Figure 3: Governing Agentic AI: Managing Autonomous Behavior and Invisible Failure in AI Systems

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From Telemetry to Evaluation

Runtime telemetry captures what AI systems are doing in production. Systematic evaluation assesses whether they are doing it well through the deliberate, repeated scoring of AI outputs against defined quality criteria. The two are complementary, and both are necessary.

Evaluation operates in two modes. Offline evaluation tests models and pipelines against curated datasets and adversarial prompt libraries before or between deployments. Online evaluation scores live production interactions using automated judges, human reviewers, or user feedback signals. For RAG-based systems, both retrieval and generation stages require independent evaluation, as a response can fail at either point, and without stage-level instrumentation, the root cause remains ambiguous (see Table 1).

Table 1: Offline vs. Online Evaluation

Dimension	Offline Evaluation	Online Evaluation
When	Before or between deployments	During live production
Purpose	Validate models and pipelines in controlled settings	Monitor real-world performance and quality
Inputs	Curated datasets, test suites, adversarial prompts	Live user interactions
Methods	Benchmarking, test scoring, adversarial testing	Automated judges, human review, user feedback
Strengths	Repeatable, controlled, safe for testing edge cases	Reflects real usage, detects emerging issues
Limitations	May not capture real-world variability	Harder to control, noisier signals

Source: Gartner

The practical implication for I&O and platform engineering leaders is that AI observability tooling selection cannot focus solely on monitoring capability (see [Market Guide for AI Evaluation and Observability Platforms](#)). Evaluation frameworks are a core component of a mature AI observability strategy, not an optional addition.

Aligning AI Performance to Business Outcomes

AI observability is not purely a technical discipline. It is the mechanism by which organizations connect AI system behavior to business value, cost management, and regulatory compliance. It is also the foundation on which a credible case for continued AI investment is built.

Outcome-based metrics for AI systems, like accuracy rates, bias reduction, and incident response times, give business leaders concrete evidence of AI's business impact. This is evidence that resonates beyond the engineering team and can withstand scrutiny at the board and executive level. Without this instrumentation, AI ROI conversations remain qualitative and difficult to defend under scrutiny.

Cost visibility is also important. AI workloads introduce consumption models that have no direct equivalent in traditional IT. Metrics like token usage, inference compute, and model API costs can scale rapidly and unpredictably. Without this visibility, AI costs are difficult to predict and harder to justify. With it, organizations can make informed decisions about model selection, deployment architecture, and usage policies.

Finally, AI observability produces the audit trails, governance evidence, and compliance documentation that regulators and enterprise risk functions increasingly require. New legal frameworks, such as the EU AI Act, NIST AI RMF and ISO 42001, place explicit obligations on organizations to demonstrate oversight and accountability over AI systems in production. Observability provides both the technical means of meeting these obligations and the documented evidence that they are being met.

Risks

Tooling Immaturity and Market Fragmentation

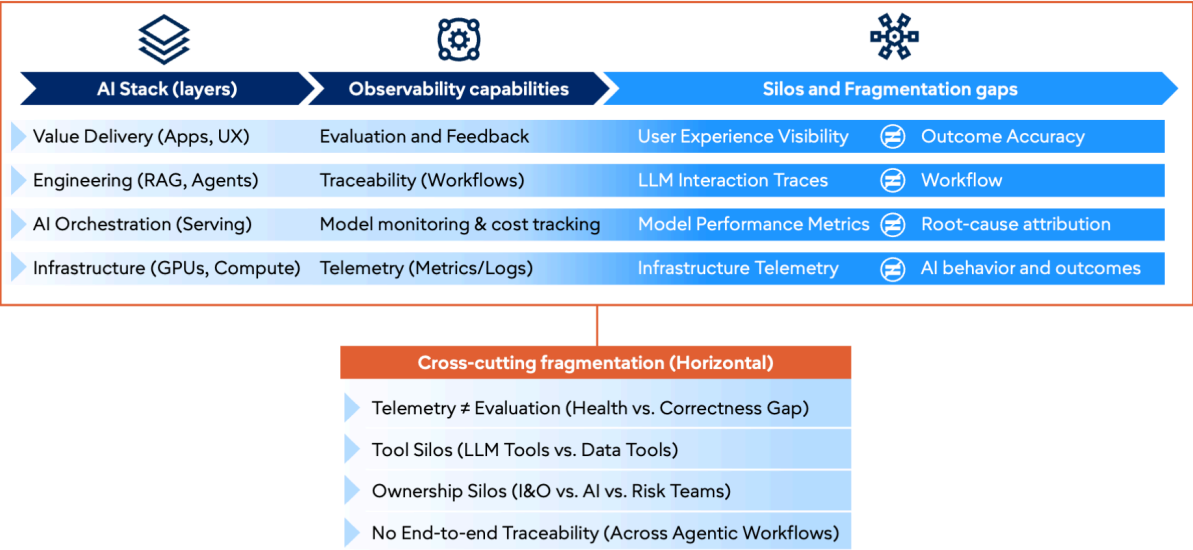
The AI observability market is growing quickly but remains immature. Specialist vendors are emerging rapidly, offering capabilities that range from LLM tracing and evaluation frameworks to bias detection and agentic workflow monitoring. However, few have reached the scale and breadth that enterprise I&O teams typically require from a strategic tooling investment.

Organizations that move too quickly risk investing in tools that are superseded or acquired before they deliver sustained value. Those that wait for the market to consolidate risk accumulating AI deployments without adequate observability coverage in the interim.

Mitigating this requires prioritizing open standards such as OpenTelemetry over proprietary instrumentation, favoring modular architectures that can accommodate tooling changes, and treating vendor selection as a decision that requires regular reassessment rather than a one-time commitment (see Figure 4).

Figure 4: AI Observability Fragmentation: Disconnected Signals Across the AI Stack

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The Governance and Ownership Gap

AI observability benefits multiple teams across the business, including I&O, data science, AI engineering, legal, compliance, and security. However, clear ownership is rarely established upfront. The result is instrumentation that reflects whoever has or had the initiative to implement it rather than a coherent organizational strategy. Critical gaps emerge at the boundaries between teams, and accountability for acting on what observability surfaces is frequently ambiguous.

This is particularly acute for agentic AI systems, where the boundary between infrastructure responsibility and model governance is genuinely unclear. When an agentic workflow produces a harmful or noncompliant output, establishing which team owns the remediation and the prevention is a governance question that tooling alone cannot answer.

Organizations should define ownership of AI observability explicitly, before instrumentation begins, rather than after gaps become visible. This means establishing clear boundaries between AI observability and traditional observability responsibilities, embedding observability requirements into AI project governance from the outset, and ensuring that legal, compliance, and security teams are active participants in strategy rather than downstream consumers of reports.

The Limits of What Observability Can See

AI observability can only observe what is instrumented and accessible. Closed third-party foundation models limit observability to inference-layer signals, with no visibility into model weights, training data, or internal reasoning. Organizations consuming AI capabilities through APIs are observing a black box from the outside.

Privacy constraints introduce further blind spots. Observability systems that capture prompts, retrieved context, and model outputs may inadvertently process personal or confidential information. This creates tension between comprehensive instrumentation and data governance obligations that require explicit resolution before observability architecture is finalized.

Shadow AI represents perhaps the most intractable limitation. AI observability cannot instrument what it does not know exists. Ungoverned AI usage requires policy and culture interventions alongside technical controls, including security and digital workplace tools. AI observability alone cannot close this gap.

Observability Debt

Organizations that do not build observability in from the start of AI deployment accumulate “observability debt.” Gaps in instrumentation become progressively harder and more expensive to close as systems grow in complexity. A model that was adequately monitored at the pilot stage may be serving millions of interactions in production one year later, with instrumentation that was never designed for that scale or scope. Architectural complexity accelerates the problem. Multimodel pipelines, RAG systems, and agentic workflows each expand the instrumentation surface, and retrofitting observability onto systems not designed for it is significantly harder than building it in from the start. The coverage gaps that result tend to stay hidden until something goes wrong.

Treating observability as a foundational requirement rather than a postdeployment addition is the most effective mitigation. This means embedding instrumentation into AI project governance from the start, establishing baseline coverage requirements before systems move to production, and scaling observability investment in step with AI deployment.

Recommendations

AI observability delivers its greatest value when it is planned deliberately, governed clearly, and connected explicitly to business outcomes. The following recommendations provide a starting point.

Start at the Outcome Layer

Start with the inference and outcome layer. This is where business impact is most tangible and where signals such as hallucination rates, output quality scores, guardrail trigger frequency, and user feedback are easiest to collect and act on. This delivers value quickly and builds organizational confidence before expanding instrumentation to the model, engineering tools, and infrastructure layers. A stepwise approach manages complexity and ensures each layer of coverage is purposeful rather than reactive.

Establish Governance and Ownership Early

Define ownership of AI observability explicitly before instrumentation begins. Establish clear boundaries between AI observability and traditional monitoring responsibilities, embed observability requirements into AI project governance from the outset, and ensure that legal, compliance, privacy, and security teams are active participants in the strategy rather than downstream consumers of reports. Without this foundation, instrumentation will reflect individual initiative rather than organizational strategy.

Evaluate Tooling for Monitoring and Evaluation Capability

Assess AI observability solutions against both runtime monitoring and systematic evaluation requirements. Prioritize open standards such as OpenTelemetry over proprietary instrumentation. Favor modular architectures and revisit vendor selection regularly as the market evolves. Do not overlook evaluation frameworks. LLM-as-judge scoring, human feedback integration, and structured testing pipelines are core components of mature AI observability, not optional additions.

Connect Observability to Business Outcomes

Define outcome-based KPIs that connect AI system performance to business value. Useful starting points include accuracy rates, bias reduction, incident response times, and user task completion. Establish granular visibility into token consumption and inference costs to make AI spend manageable and predictable. Ensure that observability produces the audit trails and governance evidence that regulators, auditors, and business stakeholders increasingly require.

Representative Providers

Arize AI, Fiddler AI, CoreWeave (Weights & Biases), LangSmith, Dynatrace, Datadog

Evidence

This research draws on insights from more than 400 interactions with Gartner clients and technology providers focused on AI observability conducted over the past 18 months.

Recommended by the Authors

Some documents may not be available as part of your current Gartner subscription.

[Introduce AI Observability to Supervise Generative AI](#)

[Predicts: AI will unlock Observability at Scale](#)

[Innovation Insight: LLM Observability](#)

[Magic Quadrant for Observability Platforms](#)

[Market Guide for Data Observability Tools](#)

[Innovation Insight for Central Observability Teams](#)

[Market Guide for Infrastructure Monitoring Tools](#)

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